

JOYOBRATA DAS

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ACADEMIC PROJECTS

Neural Netwroking From Scratch

[Github](#)

Tech Stack: Python • NumPy • Object-Oriented Programming (OOP) • Linear Algebra • Matplotlib

- Implemented a complete feedforward neural network from scratch using Python and NumPy without high-level deep learning frameworks.
- Developed Dense Layers, ReLU, Softmax, Categorical Cross-Entropy Loss, Forward & Backward Propagation, Gradient Descent, and Parameter Optimization.
- Trained the neural network on classification datasets while designing a modular, object-oriented architecture for extensibility and experimentation. (some implementations are still under development)

Crop Automation : AI-Powered Agricultural Decision Support System

[Github](#)

Tech Stack: React • Flask • Python • Google Earth Engine • Scikit-learn

- Developed a satellite-based agricultural monitoring dashboard using React, Flask, and **Google Earth Engine** for crop and stress analysis.
- Integrated **GIS visualization**, **dynamic geolocation**, and **satellite-derived vegetation** and suitability maps for **location-specific agricultural insights**.
- Implemented a **Random Forest-based crop recommendation pipeline** and designed an end-to-end decision support system for irrigation and crop monitoring.

Secure Access: Beginner Authentication System

[Github](#)

Tech Stack: Flask • Python • Bcrypt • SQLite • HTML5 • Vanilla CSS

- **Role-Based Access Control (RBAC):** Enforces access restrictions between user roles (admin vs user) on protected endpoints.
- **Brute-Force Rate Limiting:** Database-driven block lockouts implemented inside the login function (restricting users to 5 failed attempts in the last 5 minutes).
- **Audit Logging:** SQLite database tracking of successful and failed login attempts with timestamps.

Chandrayaan-3 Inspired Lunar Rover Model

[The Telegraph](#),: 19th Jan 2024

Developed and exhibited a student-built lunar rover prototype model inspired by India's space mission, showcased during EXORGIC 2024. Recognition on Telegraph

EXPERIENCE

Web Developer | School of Data Science & Forecasting, DAVV

[Website](#)

Nov 2025 – Present

- Developed and deployed the official School of Data Science & Forecasting website using PHP, HTML, CSS, and JavaScript.
- Implemented reusable components using PHP includes for better maintainability.

TECH STACK

Languages: Python, Java, C++, JavaScript, PHP, SQL

Web & Development: React, Flask, REST APIs, HTML/CSS, Vite

AI & Data Science: Machine Learning, Scikit-Learn, Pandas, NumPy, Data Analysis, Computer Vision

Tools: Git, GitHub, Linux, SQLite

EDUCATION

M.Tech Integrated Degree in Artificial Intelligence & Data Science

School of Data Science & Forecasting, DAVV, Indore | Expected Graduation: 2029

St. Joan's School (Class XI & XII)

ICSE Class X, ISC Class XI & XII

SKILLS

- Art (visit @joyobrata_05) • Graphic (visit @the_terabyteframes)
- Editing